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Trade Finance Use by Heterogeneous Firms

Francesca de Nicola¹, Alexandros Ragoussis¹

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June 1, 2026

Abstract

Letters of credit are a key trade finance instrument that covers more than 10 percent of global trade, with a notably larger role in low- and middle-income economies. Studying detailed trade data from Viet Nam, this paper documents how the use of letters of credit varies with firm characteristics. The paper shows that the probability of using a letter of credit is systematically lower for younger, smaller, and foreign-owned trading firms. Importers that are less diversified or have less trading experience are more likely to use letters of credit. Firm characteristics have the strongest effects in markets where information is scarce and enforcement is weak. These patterns are consistent with a model in which the ability to screen trading partners and the cost of bank intermediation vary with firm characteristics, and where a firm's screening ability and country institutions are substitutes. Any policy or intervention that aims at increasing the use of bank-intermediated trade finance will therefore need to take firm heterogeneity into account.

Keywords: International Trade; Trade Finance; Letter-of-credit; Firms; Development

JEL Codes: F14, F34, O16

¹ World Bank Group.

² Federal Reserve Board

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1 Introduction

Selling goods across borders is risky. If exporters deliver goods on credit, they bear the risk of non-payment by the importer. If, instead, importers pay in advance, they bear the risk of non-delivery by the exporter. This issue is especially salient for low and middle income economies, where weaker institutions and other impediments make it harder to enforce contracts. This two-sided commitment problem can be resolved with a letter of credit, a specialized financial instrument where banks monitor the trade transaction and settle the payment against a fee.

Letters of credit are widely used, and many development banks are active in the promotion of this instrument. Although global data are scarce, SWIFT data imply that letters of credit were used for about 13 percent of global trade in 2011 ([Niepmann and Schmidt-Eisenlohr, 2017a](#)). However, this share was much larger for some low and middle-income economies. China and the Republic of Korea, for example, used letters of credit for 43 and 27 percent of their imports, respectively. Viet Nam, which is the focus of this study, financed about 9 percent of its exports and 24 percent of its imports with letters of credit. Letters of credit are particularly useful at the beginning of new relationships when risks are higher, making them an important tool for internationalizing a country's production ([Antras and Foley, 2015](#); [Benguria et al., 2023](#)).

Despite their wide use, relatively little is known about the determinants of letter of credit use. Although some work has emphasized country characteristics theoretically or empirically ([Schmidt-Eisenlohr, 2013](#); [Antras and Foley, 2015](#); [Niepmann and Schmidt-Eisenlohr, 2017a](#)), the role of firm characteristics has received little attention. This is surprising, as firm heterogeneity has taken center stage in the trade literature over the last three decades.¹

To fill this gap, this paper uses a matched industrial census and customs dataset from Viet Nam, with information on the use of letters of credit, covering the universe of firms and trade flows

¹See, for example, the seminal contributions by [Bernard and Jensen \(1999\)](#), [Pavcnik \(2002\)](#), [Melitz \(2003\)](#) and [Eaton and Kortum \(2002\)](#). For a review, see, e.g., [Melitz and Redding \(2014\)](#).

from 2016-2022.

We begin by testing the relative importance of firm-level variation compared to country-product-level variation in letter of credit use through a variance decomposition. Similar to findings in the broader trade literature on export participation, firm-level variation is two to seven times more important than country-product-level variation in explaining letter of credit use.

We then study the role of specific firm characteristics. Consistent with determinants of international trade participation, we find that firm size, productivity, and multinational status also play a key role in the decision whether to use a letter of credit. Specifically, younger, smaller and foreign-owned firms are systematically less likely to use letters of credit. In addition, importers that are less diversified or have less trading experience are more likely to use letters of credit, while trade experience has no statistical effect on exporters' letter of credit use. We find that all these effects are strongest for countries with less information availability (proxied by the Vietnamese diaspora share) and countries with weaker contract enforcement. That is, using a letter of credit is determined more strongly by firms' screening abilities and access to recourse when they trade with challenging partner countries.

These patterns are consistent with a model in which the ability to screen foreign firms and the cost of bank intermediation vary with firm characteristics. The basic idea is that there is heterogeneity across firms in two dimensions. First, heterogeneity in competencies to select trading partners, manage risks, and engage in international trade without intermediation. Second, heterogeneity in the access and the cost of bank intermediation. The results suggest that cost advantages are more likely to drive the effects of age, firm size, and labor productivity, whereas differences in screening ability play a greater role for foreign ownership and trade experience. Importantly, under the assumption that firms' screening abilities and country institutions are substitutes, the model can rationalize the finding that the strength of the cost and screening effects decreases with contract enforcement and information availability in the partner country.

These new facts and theoretical insights not only contribute to a better understanding of trade finance use across firms but are also central for policy-makers who want to support trade finance for development in low and middle-income economies. Recent estimates suggest an unmet demand for trade finance in the trillions of US dollars ([Beck et al., 2023](#)). Moreover, regional studies in Africa, Asia, and the Americas suggest that improving access to and reducing the cost of bank-intermediated trade finance could increase trade by 5 to 16 percent ([IFC and WTO, 2022, 2023, 2025](#)). Because most of the variation in letter of credit use is at the firm level, firm-level heterogeneity has to be taken into account in any intervention that aims at expanding the use of letters of credit for cross-border trade, through targeted interventions to alleviate their cost and greater provision of firm advisory.

The paper contributes to several strands of the literature. It expands on the theory of payment terms in international trade ([Schmidt-Eisenlohr, 2013](#); [Antras and Foley, 2015](#); [Ahn, 2014](#); [Niepmann and Schmidt-Eisenlohr, 2017a](#); [Fischer-Thöne, 2023](#); [Garcia-Marin et al., 2022](#); [Benguria et al., 2023](#); [Kohn et al., 2024](#)) by deriving results on the effects of firm characteristics and their interactions with country variables on the use of letters of credit. To do that, it builds on a broader view of trade under imperfect contractibility in which trade is shaped by firms' ability to screen counter-parties and rely on alternative enforcement mechanisms, echoing early work on information, trust, and institutions in trade (e.g., [Greif, 1993](#); [Rauch and Trindade, 2002](#); [Araujo et al., 2016](#)). It adds to the empirical work on the determinants of trade finance in general (see, e.g., [Antras and Foley, 2015](#); [Demir and Javorcik, 2018](#)) and letters of credit specifically ([Niepmann and Schmidt-Eisenlohr, 2017a](#)) by shedding light on the firm characteristics driving letter of credit use.

In documenting these new facts, the paper also links to research that shows the adverse effects of negative letter of credit supply shocks ([Niepmann and Schmidt-Eisenlohr, 2017b](#); [Ahn and Sarmiento, 2019](#); [Kotidis et al., 2024](#)). The paper also relates to recent work on trade finance and geopolitics ([Corsetti et al., 2025](#)) by looking at the combined role of firm characteristics and country

characteristics. Finally, the analysis contributes to the wider literature on financial frictions and trade (see, e.g., [Ahn et al., 2011a](#); [Amiti and Weinstein, 2011](#); [Manova, 2013](#); [Paravisini et al., 2015](#); [Leibovici, 2021](#)), which has studied the links between financial conditions, international trade, and the macroeconomy.

The remainder of the paper is organized as follows: The next section reviews payment terms, including alternatives to letters of credit, and summarizes relevant findings from the literature. It then extends the theory of payment terms to highlight the key tradeoffs firms face and how these may vary with firm and country characteristics. Section 3 describes the data on firms, trade, and payment terms used in the empirical analysis. Section 4 presents the variance decomposition at different levels, empirical models, and results, linking observed differences to the underlying theoretical drivers of letter of credit use. The final section concludes with a discussion of policy implications.

2 Conceptual Framework, Background and Theory

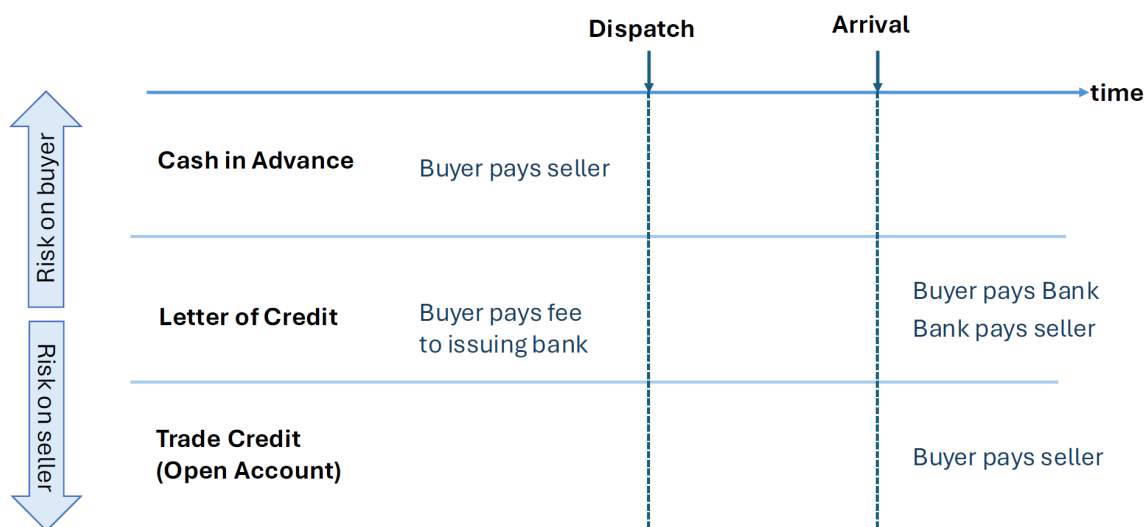
In this section, we lay out a simple model of letter of credit use. The purpose of the section is twofold. First, it provides an overview of the key payment terms and the tradeoffs faced by exporters and importers when selecting among these options. Second, it gives the reader intuition about how firm characteristics may affect the use of letters of credit and how contract enforcement and information availability at the country level may reduce these effects.

2.1 Overview of Payment Terms

Figure 1 illustrates the three most common payment terms in international trade: cash in advance, trade credit (also called “open account”), and letters of credit. With cash in advance, the importer pays before the seller sends the goods. There is a risk that the exporter does not deliver the goods after payment or delivers the goods too late, or supplies goods of inferior quality - with all these

risks being on the buyer's side. With trade credit, the exporter first delivers the goods, and there is a risk that the importer does not pay for the goods after receiving them - with all risk being on the seller's side. The third option is a letter of credit, the focus of this paper. With a letter of credit, the two commitment problems get resolved simultaneously through bank intermediation. The importer commits to paying the bank either by making a cash deposit, providing other collateral, or through a long-term relationship with the bank. The exporter only gets paid after they provide proof of delivery of the goods according to the terms that were agreed. Although letters of credit resolve a large part of the transaction risk, they tend to be costly: banks need to be compensated for the effort that goes into paperwork and monitoring, as well as for the residual risk that they may be exposed to. Trading firms, thus, face a fundamental tradeoff between, on the one hand, risks arising from lack of market information, prior experience with trading partners, or lack of recourse, and, on the other hand, the cost of bank intermediation.

Figure 1: Payment Terms in International Trade



Note: This figure illustrates the three most common payment terms in international trade.

2.2 What Do We Know about the Use of Letters of Credit?

Only a small number of papers have studied the use of letters of credit with detailed data, with most analysis having focused on the role of within-relationship dynamics and country characteristics. [Antras and Foley \(2015\)](#) and [Benguria et al. \(2023\)](#) find that letters of credit are used more for first time trade because counter-party risk declines within relationships over time due to learning about trading partners. Using SWIFT data, [Niepmann and Schmidt-Eisenlohr \(2017a\)](#) document that letters of credit are used more for larger transactions, likely due to the higher fixed cost charged by banks. In addition, they document that letters of credit are used the most for countries of intermediate risk. They rationalize this finding with a model where letters of credit are not used for the safest destinations as these are better and more cheaply served with trade credit. For the riskiest destination, letters of credit become too expensive, and exporters prefer to be paid in advance. While [Antras and Foley \(2015\)](#) and [Benguria et al. \(2023\)](#) also rely on firm-level data, they do not study the role of firm characteristics in determining the use of letters of credit.²

2.3 Formal Model

In the following, we present a simple model of payment terms that helps convey the key intuition for the tradeoffs faced by firms, and how these may be affected by firm characteristics. Consider a simplified version of the model in [Schmidt-Eisenlohr \(2013\)](#), assuming the total cost of using a letter of credit f is a function of the payment value P^{LC} and a fixed component m : $f = fP^{LC} + m$. In the model a risk-neutral exporter and a risk-neutral importer are matched. The exporter has all the bargaining power and makes a take-it-or-leave-it offer to the importer. There are three payment terms: cash in advance, trade credit, and letters of credit.

²In a recent update, [Benguria et al. \(2023\)](#) study how trade credit dynamics vary with relationship age across firms with different sizes and risk profiles. However, their analysis focuses on the tradeoff between trade credit and cash in advance, and it does not study the role of firm heterogeneity beyond its effects on learning dynamics.

2.3.1 Payment Terms

Trade Credit With trade credit, the exporter first delivers the goods and then gets paid by the importer. However, some importers are reliable (fraction $\eta^I \in [0, 1]$) while others are not (fraction $1 - \eta^I$). If an importer is unreliable they have an opportunity to divert funds with probability $1 - \phi^I$, where $\phi^I \in [0, 1]$ reflects country characteristics such as the quality of rule of law or information frictions enabling deceit. The exporter maximizes the following expected profits $\Pi^{TC,E}$ subject to the participation constraint of the importer³

$$\begin{aligned} \mathbb{E}[\Pi^{TC,E}] &= \frac{\tilde{\eta}^I}{1 + r^E} P^{TC} - C, \\ \text{s.t. } \mathbb{E}[\Pi^{TC,I}] &= R - P^{TC} \geq 0. \end{aligned}$$

where C is the exporter's cost of production, R importer's revenue from selling the goods to final consumers, r^E and r^I interest rates at the origin and destination country respectively and $\tilde{\eta}^I = \eta^I + (1 - \eta^I)\phi^I$ is the probability that the importer pays after receiving the goods. A payment happens if the importer is reliable (η^I) or if the importer is unreliable ($1 - \eta^I$) but does not have a diversion opportunity (ϕ^I). The optimal payment P^{TC} makes the importer indifferent by setting $P^{TC} = R$, resulting in expected profits of

$$\mathbb{E}[\Pi^{TC,E}] = \frac{\tilde{\eta}^I}{1 + r^E} R - C.$$

Cash in Advance With cash in advance, the importer first pays for the goods and then the exporter delivers them. A delivery takes place with probability $\tilde{\eta}^E = \eta^E + (1 - \eta^E)\phi^E$. A reliable

³Under relatively weak assumptions, the optimal solution is a pooling case where unreliable importers or exporters always want to imitate reliable firms. Throughout the paper, we assume that these conditions are met and we can focus on maximizing the problem of reliable firms. For details on the conditions for pooling and separating cases, see [Schmidt-Eisenlohr \(2013\)](#) and [Garcia-Marin et al. \(2022\)](#).

exporter maximizes expected profits

$$\begin{aligned} \mathbb{E}[\Pi^{CIA,E}] &= P^{CIA} - C, \\ \text{s.t. } \mathbb{E}[\Pi^{CIA,I}] &= \frac{\tilde{\eta}^E}{1+r^I} R - P^{CIA} \geq 0. \end{aligned}$$

The optimal payment that makes the importer indifferent is $P^{CIA} = \frac{\tilde{\eta}^E}{1+r^I} R$, delivering expected exporter profits of

$$\mathbb{E}[\Pi^{CIA,E}] = \frac{\tilde{\eta}^E}{1+r^I} R - C.$$

Letter of Credit With a letter of credit, a bank guarantees payment, subject to meeting contractual conditions, upon receipt of the shipping documents. At the same time, the importer only receives goods after committing to the bank to pay for the imports. We assume that the bank is able to resolve all commitment problems on both sides but requires a fee.⁴ The fee has a fixed cost component m and a component that is proportional to the payment amount, so that total costs of using a letter of credit are given by $f = f^{PLC} + m$.⁵ The exporter now maximizes

$$\begin{aligned} \mathbb{E}[\Pi^{LC,E}] &= \frac{P^{LC}}{1+r^E} - C, \\ \text{s.t. } \mathbb{E}[\Pi^{LC,I}] &= \frac{R - P^{LC}}{1+r^I} - (f^{PLC} + m) \geq 0. \end{aligned}$$

⁴It could be the case that banks are still exposed to residual risk. We abstract from this issue here. See [Niepmann and Schmidt-Eisenlohr \(2017a\)](#) for details on how to expand the model in this direction.

⁵For simplicity, the cost is assumed to be entirely borne by the importer, although exporters also face (typically smaller) so-called negotiation fees that can be variable or flat, and optionally a confirmation fee, involving additional guarantees if contractual terms are not met. In addition, fixed costs incurred by importing firms are in reality often non-monetary and reflect the time and effort for issuing a letter of credit: importers' banks typically only charge a variable component proportional to the payment value. In addition, the costs of a letter of credit can also be thought of as a reduced form way of capturing rationing, as small and medium size firms' letter of credit applications in developing economies often get rejected by banks.

Solving for the optimal payment delivers $P^{LC} = \frac{R-(1+r^I)m}{1+f(1+r^I)}$, resulting in expected exporter profits of

$$E[\Pi^{LC,E}] = \frac{R - (1 + r^I)m}{(1 + r^E)(1 + f(1 + r^I))} - C.$$

2.3.2 Comparative Statics

Baseline Comparison To study the attractiveness of a letter of credit, we must compare it to the best available alternative. Let $E[\Pi^{Alt}]$ denote the expected profit from the best alternative, defined as the envelope of profits from trade credit and cash in advance

$$E[\Pi^{Alt}] = \max \{E[\Pi^{TC}], E[\Pi^{CIA}]\}.$$

We define the expected net gain from using a letter of credit as $E[\Delta\Pi] = E[\Pi^{LC}] - E[\Pi^{Alt}]$. Plugging in the earlier expressions delivers

$$E[\Delta\Pi] = \frac{R - (1 + r^I)m}{(1 + r^E)(1 + f(1 + r^I))} - \max \left\{ \frac{\tilde{\eta}^I R}{1 + r^E}, \frac{\tilde{\eta}^E R}{1 + r^I} \right\} \quad (1)$$

We can decompose this net gain into two distinct components: a *Cost Effect*, which negatively affects letter of credit use because letters of credit are more expensive compared to their alternatives; and a *Risk Effect*, which positively affects letter of credit use because letters of credit reduce transaction risk relative to their alternatives.

$$E[\Delta\Pi] = \underbrace{\frac{-(1 + r^I)m}{(1 + r^E)(1 + f(1 + r^I))}}_{\text{Cost Effect} < 0} + \underbrace{\frac{R}{(1 + r^E)(1 + f(1 + r^I))} - \max \left\{ \frac{\tilde{\eta}^I R}{1 + r^E}, \frac{\tilde{\eta}^E R}{1 + r^I} \right\}}_{\text{Risk Effect}} \quad (2)$$

The cost effect is negative, because the monitoring fee of a letter of credit reduces profits for firms using it relative to those that do not. For small enough variable cost f , the risk effect is

positive because the letter of credit increases the probability of payment, and thus the expected profitability of the transactions. The strength of the risk effect depends on the screening abilities and the underlying risk of the counter-party. The riskier the firms and the less good they are at screening (low $\tilde{\eta}$), the larger the benefit from using a letter of credit.

Proposition 1 (Letter of Credit Determinants). *The expected net gain from using a letter of credit relative to the best alternative*

- (i) *decreases with the monitoring cost, m , and the variable letter-of-credit cost, f^{LC} ;*
- (ii) *weakly decreases with the probability of successful direct payment or delivery without intermediation ($\tilde{\eta}^I$ or $\tilde{\eta}^E$);*
- (iii) *measured per unit of transaction value, increases with revenues, R , because the fixed monitoring cost is spread over a larger transaction.*

Proof. See appendix A. □

Proposition 1 summarizes the key determinants of the payment choice between letters of credit and their alternatives. First, letters of credit are less attractive when monitoring or variable intermediation costs are higher. Second, when the relevant non-intermediated alternative becomes safer, the relative benefit of a letter of credit falls. Third, because letters of credit involve a fixed cost component, larger transactions are more likely to justify their use: scale makes it easier to absorb the fixed cost of bank intermediation.

2.3.3 The Role of Firm Characteristics

How do firm characteristics affect the use of letters of credit? Assume that a firm characteristic (e.g., size, trade experience, or ownership) is captured by parameter $\theta^i \in [0, 1]$ with $i \in \{I, E\}$. Assume further that, due to better screening capabilities, exporter firm quality raises the probability of

being matched with a reliable importer, while importer firm quality raises the probability of being matched with a reliable exporter. Formally, let $\eta^j = \theta^i$, where $i \neq j$. Then the probability of payment or delivery is

$$\tilde{\eta}^j = \theta^i + (1 - \theta^i)\phi^j, \quad (3)$$

for $i, j \in \{I, E\}$ with $i \neq j$. That is, a payment or delivery only fails if both firm i 's screening and the institutions in country j fail.

Similarly, assume that the bank's monitoring cost, m , declines with both firm and country quality. Specifically, let $\bar{m}$ be the base cost of monitoring in a zero-information environment and assume that the monitoring costs are given by

$$m(\theta^E, \theta^I, \phi^E, \phi^I) = \bar{m} [(1 - \theta^E)(1 - \phi^I) + (1 - \theta^I)(1 - \phi^E)]. \quad (4)$$

The monitoring cost has two components, one for each side of the transaction. On each side, the bank must verify that the counterparty will fulfill its obligation. This verification is easier when the firm on that side of the transaction is better at screening its trading partner and when the counterparty operates in a country with strong institutions. For instance, the bank's cost of verifying that the importer will pay depends on how well the exporter has vetted the importer (θ^E) and on the strength of contract enforcement in the importer's country (ϕ^I). Symmetrically, the cost of verifying that the exporter will deliver depends on the importer's screening ability (θ^I) and the exporter's country institutions (ϕ^E).

For tractability, we focus on the case where there are no variable costs for a letter of credit ($f = 0$). As before, the expected net gain from using a letter of credit relative to the best alternative is

$$E[\Delta\Pi] = \frac{R - (1 + r^I)m}{1 + r^E} - \max \left\{ \frac{\tilde{\eta}^I R}{1 + r^E}, \frac{\tilde{\eta}^E R}{1 + r^I} \right\}. \quad (5)$$

Consider the effect of exporter firm quality θ^E on letter of credit use. Because each firm characteristic enters only one component of the monitoring cost and at most one $\tilde{\eta}$, the derivative of $E[\Delta\Pi]$ with respect to θ^E takes a simple form:

$$\frac{\partial E[\Delta\Pi]}{\partial \theta^E} = \underbrace{\frac{(1+r^I)\bar{m}(1-\phi^I)}{1+r^E}}_{\text{Cost channel} > 0} - \underbrace{\frac{(1-\phi^I)R}{1+r^E}}_{\text{Screening channel} \geq 0} \cdot \mathbf{1}_{[\text{TC}]}, \quad (6)$$

where the screening channel is only active in the region where TC is the best alternative, that is $\mathbf{1}_{[\text{TC}]} = 1$. When trade credit is the best alternative, the sign is ambiguous: a better exporter reduces the bank's monitoring cost (positive) but also makes trade credit safer by improving the screening of the importer (negative). When cash in advance is the best alternative, the exporter's screening ability is irrelevant for the risk of the alternative—since the importer pays first, it is the exporter's delivery risk that matters—so only the cost channel operates, and the effect is unambiguously positive.

The net effect on letter of credit use from an increase in firm quality thus depends on whether the cost effect or the risk effect dominates. If the reduction in monitoring costs exceeds the improvement in screening, the firm characteristic will have a positive effect on letter of credit use (Cost Dominance). Conversely, if the improvement in screening capabilities exceeds the reduction in banking costs, the firm characteristic will have a negative effect on letter of credit use (Screening Dominance). Which of these two effects dominates may be different for every firm characteristic like size, foreign ownership status, etc.

2.3.4 Interactions with Country Characteristics

We now turn to how country-level institutional quality interacts with firm characteristics. In both regions of the best alternative, the main effect of exporter firm quality in (6) is proportional to

$(1 - \phi^I)$. The cross-derivative with importer-country institutions ϕ^I is

$$\frac{\partial^2 E[\Delta\Pi]}{\partial\theta^E\partial\phi^I} = \begin{cases} \frac{R - (1 + r^I)\bar{m}}{1 + r^E} & \text{if trade credit is the best alternative} \\ -\frac{(1 + r^I)\bar{m}}{1 + r^E} & \text{if cash in advance is the best alternative} \end{cases} \quad (7)$$

In both cases, the cross-derivative has the opposite sign of the main effect. As importer-country institutions improve ($\phi^I \rightarrow 1$), the factor $(1 - \phi^I)$ shrinks, compressing the marginal effect of exporter firm quality toward zero. By a symmetric argument, the main effect of importer firm quality θ^I is proportional to $(1 - \phi^E)$, and the cross-derivative with exporter-country institutions ϕ^E always has the opposite sign of the main effect. This result is summarized in the following proposition.

Proposition 2 (Firm Characteristics and Letter of Credit Use). *Assume that the monitoring cost is given by (4) and that $f = 0$. Then:*

- (i) *A firm characteristic that primarily reduces the cost of bank intermediation has a positive effect on letter of credit use (Cost Dominance), while a firm characteristic that primarily improves screening has a negative effect (Screening Dominance).*
- (ii) *Better partner-country institutional quality attenuates the effect of firm characteristics on letter of credit use: the cross-derivative of the net gain with respect to firm quality and partner-country institutions always has the opposite sign of the main effect.*
- (iii) *As partner-country institutions approach perfection ($\phi \rightarrow 1$), the marginal effect of firm characteristics shrinks to zero, compressing heterogeneity in letter of credit use across firms.*

Proof. See Appendix A. □

The intuition behind the attenuation result is straightforward. Both the risk of the alternative payment method and the bank's monitoring cost depend on the same two factors: firm screening

quality and partner-country institutional quality. As partner-country institutions improve, they substitute for firm-level screening, reducing the risk differential between high- and low-quality firms under the alternative payment terms. At the same time, they ease the bank’s verification task, leaving less room for firm quality to reduce monitoring costs further. Through both channels, better partner-country institutions compress the differences in letter of credit use across firms.

In the empirical analysis that follows, we test Proposition 2 by interacting firm characteristics with measures of partner-country information availability, exploiting cross-country variation in the Vietnamese diaspora share as a proxy for information about the trading partner’s country and the World Bank’s rule of law index as a proxy for contract enforcement.

3 Data and Basic Patterns

Viet Nam has been one of the most internationally integrated economies in the world: the country’s trade-to-GDP ratio was at 185 percent in 2022, several times higher than the global average of 62 percent (IFC and WTO, 2023). Although fairly diversified in terms of products and geographies, the structure of Viet Nam’s imports and exports has undergone significant changes in the past decade. China and the United States have grown to become the main export destinations, jointly accounting for over 40 percent of the country’s total trade, while exports to other regions have been growing at a slower pace. The growth of Viet Nam’s electronic industry has been leading the way, driven primarily by foreign investors. In 2021, specifically, multinational enterprises contributed to around 70 percent of Viet Nam’s total trade flows. Large and mega firms contributed more than 80 percent of exports from the country. The country relies significantly on the import of intermediate goods and inputs (60 percent of total in 2020-2022) which tend more frequently to require letters of credit (IFC and WTO, 2022, 2023).

3.1 Customs and Census

Our empirical analysis is based on detailed international trade and firm-level data provided by Vietnamese Customs and the General Statistics Office.

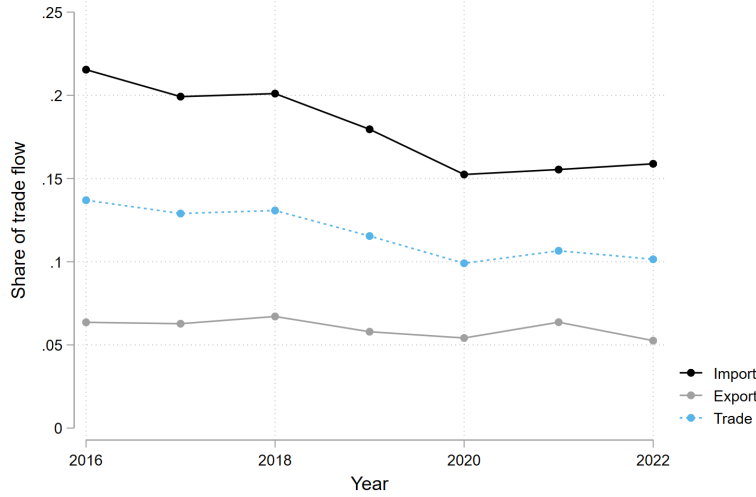
Firm characteristics. We obtain firm characteristics from the Viet Nam Enterprise Survey (VES) by the General Statistics Office (GSO). Our data includes annual information for about 517,000 to 756,000 formal firms between 2016 and 2022. We observe basic firm information (e.g., year of establishment, 5-digit sector, location at the commune level, foreign and state ownership) as well as yearly balance sheet and income statement items, such as employment, revenues, costs, and total fixed capital. Since 2016, VES data is integrated with data from the tax administration, so our effective sample is the census of all active⁶ formal firms with a tax ID. We merge the VES with Custom data using firms' tax IDs.

Trade flows and letter of credit use. We obtain firm-level export and import records from Vietnamese Customs. For each firm, the data includes the value of export and import transactions aggregated at the monthly level, by partner country, and by HS4 product. In this paper, we focus on transactions occurring between 2016 and 2022, during which firm-level data is also available. We keep firms with at least 5 employees to mitigate potential under-reporting and sample selection issues with very small firms. In total, the data cover 36,596 unique exporters, 100,071 unique importers, and 241 partner markets.

Importantly, for each firm-month-product-partner, we know both the total value of cross-border payments as well as the breakdown by different payment modes, since Vietnamese firms are required to report them in customs' declarations. The data covers 12 payment choices, including bank-intermediated trade finance instruments such as letter of credit, documentary collection (document against acceptance, document against payment), and other financing terms that fall under cash or

⁶Defined as firms that had some positive revenue or operating costs within the last 2 years.

Figure 2: Share of Letters of Credit in Total Trade Flows



Source: Matched Customs and Industrial Census (VES) data.

Note: the sample only includes trade transactions for firms with a least 5 employees.

credit. We focus on the use of letter of credit which is the most consistently reported payment choice in the data.⁷ Figure 2 plots the share of imports, exports and total trade of Viet Nam covered by letters of credit between 2016 and 2022. Importers are more likely to use letters of credit than exporters: between 15-22 percent of imports rely on a letter credit, while only 5-6 percent of exports do so. There has been a notable decrease in letter of credit use by importers over time, consistent with global trend reported in the previous literature ([Committee on the Global Financial System, 2014](#); [Ganesh et al., 2018](#)).

⁷The 12 payment choices available in the data include: Document against acceptance (DA), Document against payment (DP), Cash against documents (CAD), Offsets of debts, Cash payment, Cheque payment, Capital contribution, Barter, Debt repayment, Bill of exchange, Temporary transactions for good repairs (KhongTT), Letter of Credit (LC), Joint venture capital contribution, Open Account (OA), Telegraphic transfer reimbursement (TTR), and Other. We exclude "KhongTT" to avoid double counting of trade flows. Another reason for our focus on letter of credit is due to potential misreporting in the Customs data. Unfortunately, there seems to be a misclassification in the data where an implausibly large share of firms reported using TTR for their trade transactions in earlier years (accounting for about 70%-80% of trade in 2016 and 2017). Further, firms increasingly reported "Other" as their payment method, due to a change in instructions by Custom. From 2016-2022, "TTR" and "Other" consistently cover more than 70% of import value and more than 80% of export value. Information on the use of letters of credit, in contrast, is consistent with previous estimates in [Niepmann and Schmidt-Eisenlohr \(2017a\)](#), who report that in 2013 9.1 percent of Vietnamese exports and 24.4 percent of Vietnamese imports were done with letters of credit.

3.2 How Different Are Trade Finance Users?

In the following, we will compare four groups of firms: all firms (including non-traders), exporters, importers, and firms trading with letters of credit. We first discuss how, in line with earlier literature, exporters and importers differ from non-traders. We then show that there is an additional layer of firm heterogeneity when moving from general exporters and importers to those firms that trade internationally with letters of credit.

Table 1: Descriptive Statistics: Means and Standard Deviations by firm group. 2016-2022

	All	Importers	Importer (LC)	Exporters	Exporter (LC)
Age	8.157 (6.146)	9.912 (7.174)	13.263 (8.463)	10.834 (7.924)	12.661 (10.839)
Employment	29.424 (144.226)	97.398 (362.694)	132.306 (347.539)	178.294 (503.425)	216.094 (484.102)
Foreign-owned	0.049 (0.216)	0.236 (0.425)	0.114 (0.317)	0.353 (0.478)	0.220 (0.414)
State-owned	0.007 (0.083)	0.009 (0.097)	0.030 (0.172)	0.013 (0.115)	0.027 (0.162)
Productivity (log Sales/Employee)	5.961 (1.731)	6.894 (1.391)	7.595 (1.280)	6.711 (1.421)	7.123 (1.426)
Distance to international port	0.813 (0.767)	0.525 (0.545)	0.561 (0.561)	0.556 (0.615)	0.704 (0.708)
Years of experience in imports		3.148 (2.638)	3.953 (2.667)		
Number of transactions (Import months)		2.489 (1.779)	2.780 (1.713)		
Number of products imported		11.134 (19.265)	13.347 (21.795)		
Number of countries (Imports)		3.747 (4.937)	5.626 (6.116)		
Years of experience in exports				3.384 (2.799)	4.266 (2.721)
Number of transactions (Export months)				3.795 (2.688)	4.397 (2.358)
Number of products exported				4.359 (7.646)	4.560 (6.940)
Number of countries (Exports)				5.026 (8.057)	9.083 (10.947)
Observations	2110518	315878	50162	150529	22713

Source: Matched Customs and Industrial Census (VES) data.

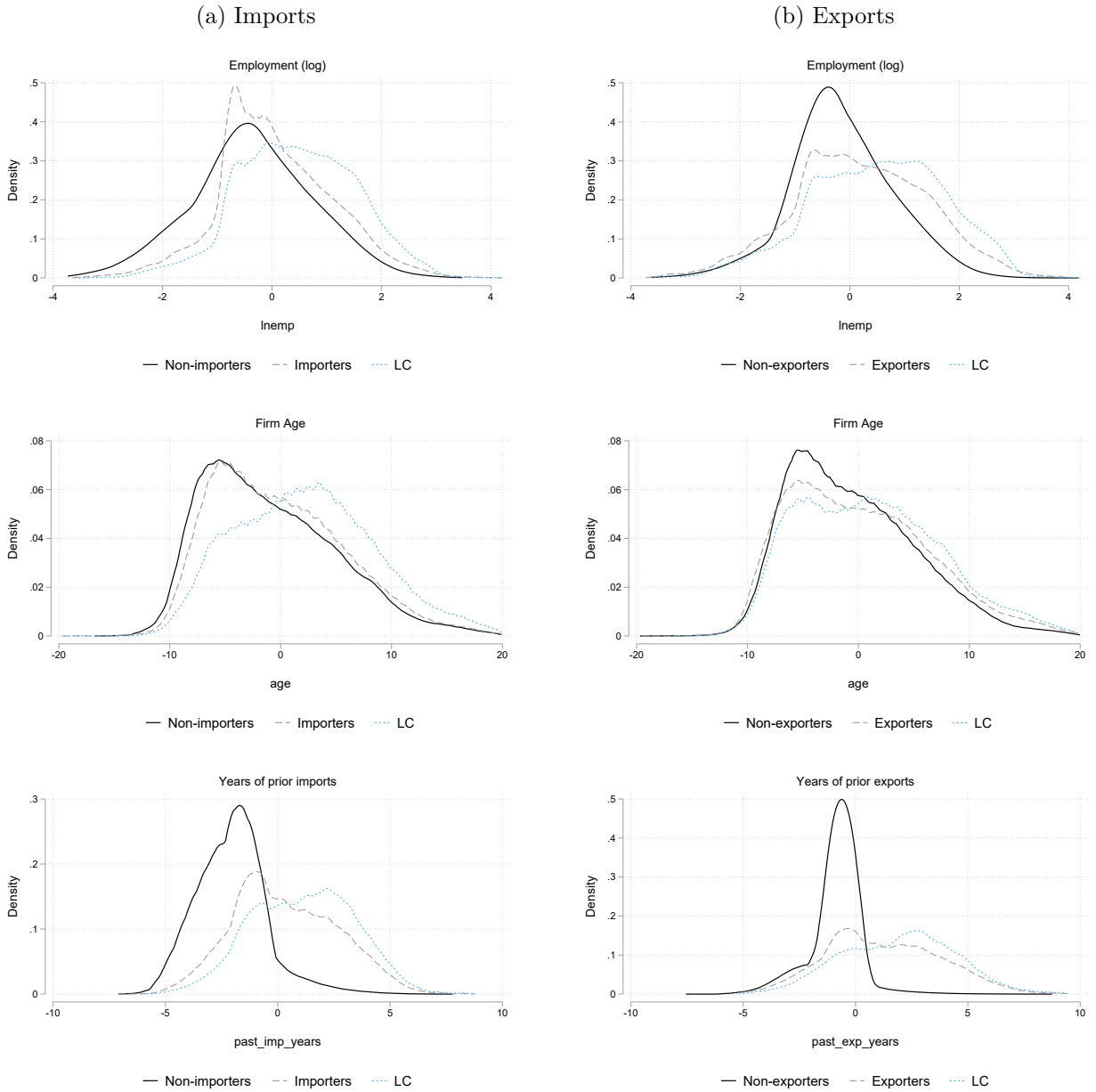
Note: Means are reported on top and Standard Deviations at the bottom in the parenthesis. The sample retained for these statistics includes firms with a least 5 employees.

We start with simple summary statistics which are reported in Table 1. Confirming well-established results in the international trade literature, we find that both exporters and importers tend to be larger, older, and more likely to be foreign-owned. Exporters and importers also have higher labor productivity levels than non-trading firms and are, on average, located closer to international ports. In addition, exporters and importers in Viet Nam are more likely to be state-owned.

We next move to the new facts about firms that engage in international trade using letters of credit. On average, letter of credit users are considerably larger, older, more productive, and have more years of trade experience compared to other traders — whether importers or exporters. They engage in multiple transactions over a greater number of months per product-country pair and trade with a wider number of countries and products every year. Letter of credit users are less likely to be foreign-owned, more likely to be state-owned, and are located farther from international ports than the average trader. That is, while some characteristics like size and productivity are even more pronounced for firms using letters of credit, some other findings, like those on multinational status and distance to a port, point to a complex relationship between firm characteristics and trade finance use.

Letters of credit are used more intensively in the international trade of certain products (Crozet et al., 2022), and by extension, are more common in some industries than others. To control for these industry-level preferences in letter of credit use, we demean key firm characteristics by industry-year and examine their distribution across the different groups of firms. Figure 3 shows the results, revealing that the likelihood of using a letter of credit increases with firm size also within an industry. Letter of credit use also increases with firm age and years of trading experience.

Figure 3: Larger, older, more experienced trading firms tend to use letters of credit more often



Source: Matched Customs and Industrial Census (VES) data.

Note: Employment is logged and demeaned at the 2-digit ISIC sector level. Kernel density estimation bandwidth = 0.5. The sample retained for these statistics includes firms with a least 5 employees.

4 Results

This section presents our main empirical results. It first presents evidence from variance decompositions on the role of firm characteristics in explaining letter of credit use. It then presents our main empirical specifications, followed by the main results, evidence on heterogeneity and validation checks.

4.1 What Factors Explain Variation in Use of Letters of Credit?

Variance decomposition. The literature on bank-intermediated trade finance has primarily focused on product and country characteristics to explain the use of letters of credit ([Antras and Foley, 2015](#); [Niepmann and Schmidt-Eisenlohr, 2017a](#); [Crozet et al., 2022](#)). How important are these characteristics compared to firm-level variation? To answer this question, we run an ANOVA decomposition, as reported in table 2. It turns out that product and country characteristics account for only a small portion of the variation in the use of letters of credit. In contrast, firm-level characteristics explain a significantly greater share of the variation. This is particularly evident for import transactions, where firm-year variation accounts for over 38 percent of the total variation in letter of credit use, compared to just 5.5 percent explained by product-partner-year variation. For export transactions this difference is smaller. Still, for exports, firm-year variation has more than twice the explanatory power than product-partner-country level variation (25 percent versus 11.5 percent).

Observable firm characteristics—such as size, sector of activity, age, and experience in trade—account, in turn, for only a portion of the firm-year variation in the use of letters of credit. Methods for disentangling the individual contribution of each characteristic to the total variation are inherently imprecise.⁸ Rather than emphasizing the precise amount of variation explained by different decom-

⁸Reporting the residual R-squared explained by the addition of a single variable to a regression, in a Type I decomposition, depends on the ordering of variables and tends to ignore interrelations among them.

Table 2: Variance Decomposition. ANOVA 2016-2022

	Exporter LC	Importer LC
Model 1		
Firm	23.46	34.01
Partner	1.82	0.21
Product	0.71	2.34
Year	0.08	0.07
Model 2		
Firm x Year	28.15	40.73
Partner x Year	1.85	0.22
Product x Year	0.81	2.27
Model 3		
Firm x Year	25.02	37.45
Country x Product x Year	11.58	5.63

Source: Matched Customs and Industrial Census (VES) data.

positions (such as Type I or Shapley Decomposition), in the next section we examine the relative importance of firm characteristics by standardizing variables and examining the effect of deviations along the distribution on the probability of using a letter of credit.

4.2 Empirical Specifications

Baseline specification. We now turn to our baseline specification to assess the role of firm characteristics in the use of letters of credit in imports and exports. We consider firm characteristics including age, state or foreign ownership, employment, labor productivity, and experience with international trade measured as the number of years since 2012 that a firm has been trading internationally and the number of product markets that firms are currently engaged in. These observable characteristics may affect both the cost of bank intermediation and firm’s ability to reduce counter party risk (screening capabilities). For example, older, larger and foreign firms may have better access to market intelligence or physical presence in multiple countries, enabling them to more effectively select suppliers and clients, and to enforce commercial agreements across borders, captured by a higher θ in our theoretical framework. They may, at the same time, be

considered less risky by banks and have established financial relationships involving lower cost of obtaining or confirming letters of credit (specifically m and f in theory). The balance between these two effects in the choice of letters of credit is ultimately an empirical question. We thus estimate the following equation:

$$LC_{ipjt} = \beta_0 + \beta_1 X_{it} + \beta_2 Z_{ipjt} + \gamma_{pjt} + \lambda_{st} + \phi_{rt} + \varepsilon_{ipjt}, \quad (8)$$

where LC_{ipjt} is the probability of firm i using a letter of credit for trade of product p with country j in year t . X_{it} is a vector of firm i characteristics in year t , Z_{ipjt} is a vector of the firm's trade activity, consisting of the total dollar value of trade and the number of months with customs transactions of firm i in product p with country j , allowing insights on the theoretical predictions regarding the impact of revenue R and monitoring fixed cost m of letters of credit (presumed to decrease with number of same product-country transactions). The extensive set of fixed effects includes product-partner-year effects, γ_{pjt} , sector-year effects, λ_{st} , and province-year effects ϕ_{rt} . Standard errors are clustered at the firm level.

The specification adds to the existing literature by focusing on firm-level variables. Product-partner-year and sector-year fixed effects eliminate variation along dimensions explored in previous studies across partner countries (Niepmann and Schmidt-Eisenlohr, 2017a) and products traded (Crozet et al., 2022), while province-year fixed effects account for local availability of trade finance products.⁹

Selection into international trade. Only firms that are already engaged in trade make use of letters of credit. As a result, estimates of the effects of firm characteristics on the use of letters of

⁹Not focusing on letters of credit specifically, several studies have highlighted the role of country-level financial conditions in shaping trade flows and trade finance (Berman and Héricourt (2010), ?, Auboin and Engemann (2014), Ahn et al. (2011b), Chor and Manova (2012)), as well as the influence of institutions, regulations, and geographic distance between trading partners (Costello (2019), Brandi and Schmitz (2015)). Market and supplier competition have also been shown to be important determinants of trade finance more broadly defined (Demir and Javorcik (2018); Chod et al. (2019))

credit are conditional on trading status and do not extend to potential traders. To assess how the decision to use letters of credit could have varied with firm characteristics across the entire firm population, it is therefore necessary to account for selection into exporting or importing.

Specifically, sample selection bias can occur if unobserved factors — such as managerial quality, access to market intelligence, or cost advantages in bank intermediation — influence both the likelihood of trading and the use of trade finance. While this can be partially controlled through fixed effects, other unobservable factors may vary over time, leading to a downward bias on the estimated effect of firm characteristics over the full population of firms including currently non-trading firms.

To mitigate this bias, we implement the Heckman correction ([Heckman, 1979](#)) using a two-step procedure. In the first stage (selection equation), we estimate a probit regression of the probability to export or import on the same set of firm characteristics as in our baseline, using the entire industrial census firm-year level sample. As exogenous variation for the first stage, we use the distance to ports measured at the district level (the 3rd administrative level in Viet Nam). This variable is correlated with the propensity of firms to trade ([Gries et al., 2009](#)), but arguably unrelated to the use of trade finance. We calculate Inverse Mills Ratios and include them in the outcome equation, which is estimated over the sample of trading firms at the firm-product-partner-year level. We bootstrap standard errors following [Cameron et al. \(2008\)](#).

4.3 Main Results

Tables [3](#) and [4](#) present our main results for importers and exporters, respectively. The dependent variable used is binary indicating whether a letter of credit has been used, and rescaled to 100 (see Appendix [B](#) for validation using share of trade covered by letter of credit as a continuous dependent variable). The baseline specification with only firm characteristics and fixed effects is reported in column 1. In column 2, we control additionally for trade characteristics including the total trade

value and the frequency of trade transactions proxied by number of months in a year when there were transactions recorded for that particular product-country pair. Column 3 adds a control for selection into trade: we report second-stage results including the Inverse Mills Ratio from the first stage of the Heckman estimation. Finally, in column 4, we include firms fixed effects to control for time-invariant unobservable characteristics at the firm level.

Letters of credit for imports. Results on letter-of-credit use for imports are shown in Table 3. First, in all specifications, letter of credit use is positively correlated with the value of the transaction and negatively correlated with the number of transactions, consistent with the prediction that letter of credit use decreases in monitoring cost and increases in revenue. Results on firm characteristics are qualitatively and quantitatively similar across the specifications shown in columns 1 and 2. The use of letter of credit by importers is positively correlated with firm age, state ownership, firm size and revenues per worker. In contrast, it is negatively correlated with foreign ownership and trade experience.

Following the intuition laid out in theory, the results suggest that cost advantages are more likely to drive the effects of age, firm size, state ownership, and labor productivity, whereas differences in screening ability may play a greater role for foreign ownership and import experience, as proxied by the number of years with importing experience and the number of products and markets. In the case of foreign ownership screening advantages are likely highest, as intra-firm trade between affiliates involves no information friction reducing the need for letters of credit to a minimum.

The quantitative relevance of firm-level variables varies. Changes in several firm characteristics appear to have a small effect on the probability of using a letter of credit for imports: An additional year of operations, and one additional product-partner relationship each increase the probability of using letters of credit by less than 1 percent, while a unit increase in firm size (in logarithmic scale), 1 million USD of additional revenue per worker, and prior trade experience by 1-2 percentage

Table 3: Linear Probability Results: Importers 2016-2022

	(1)	(2)	(3)	(4)
Age	0.058** (0.023)	0.058** (0.023)	0.050* (0.026)	
Foreign ownership	-6.988*** (0.257)	-6.685*** (0.262)	-13.387*** (0.954)	0.239 (0.260)
State ownership	6.145*** (1.945)	6.188*** (1.956)	6.893*** (2.033)	0.086 (0.471)
Employment (log)	1.201*** (0.081)	0.983*** (0.080)	-0.817*** (0.172)	0.066 (0.086)
Revenue per worker	1.491*** (0.074)	1.239*** (0.071)	0.140 (0.133)	0.170*** (0.059)
Imported for 1+ years	-1.718*** (0.180)	-1.472*** (0.177)	-1.785*** (0.243)	-0.483*** (0.119)
Number of country-products imported	-0.011*** (0.001)	-0.008*** (0.001)	-0.004*** (0.001)	0.002* (0.001)
Value of transactions (log Imports)		0.905*** (0.022)	0.930*** (0.018)	0.670*** (0.015)
Number of transaction months (imports)		-0.265*** (0.026)	-0.258*** (0.031)	-0.087*** (0.013)
Inverse Mills Ratio			-10.615*** (1.141)	
Product-Partner-Year FE	Yes	Yes	Yes	Yes
Sector-Year FE	Yes	Yes	Yes	Yes
Province-Year FE	Yes	Yes	Yes	Yes
Firm FE	No	No	No	Yes
R ²	0.26	0.26	0.27	0.53
Observations	3,320,731	3,320,731	2,904,608	3,309,628

Source: Matched Customs and Industrial Census (VES) data.

Note: The sample retained for these statistics includes transactions of firms with at least 5 employees, aggregated at the firm-product-partner-year level. The dependent variable on letter of credit use is binary, rescaled to 100 for positive use. Errors in all specifications are clustered at the firm level. Specification (3) errors are bootstrapped following the introduction of Inverse Mills Ratio as a regressor. Standard errors in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels respectively.

points.¹⁰ The correlation between the number of products and countries a firm imports from and the use of letters of credit is significantly negative, but small in magnitude. By contrast, state-

¹⁰We find evidence of non-linear effects for firms' age: they are statistically significant, but economically unimportant and are thus omitted in Table 3.

owned firms are 6 to 7 percentage points more likely to use letters of credit than private firms, while foreign-owned firms are 7 to 13 percentage points less likely to use letters of credit than domestic firms. The intensity of letter-of-credit use by state-owned trading firms may be associated with the fact that the banking sector in the country is dominated by state-owned commercial banks and joint ventures,¹¹ involving preferential access or arrangements beyond pricing of letters of credit.

Once we account for unobservable characteristics associated with the decision to trade (see column 3 versus column 2), the magnitude and significance of the effects of revenue per worker, and, to a lesser extent, firm age decline. These changes in coefficients likely reflect that non-trading firms, on average, have lower capabilities for engaging in foreign markets or face higher barriers to bank intermediation than trading firms. In other words, advantages related to age, size, or productivity are less pronounced among trading firms due to selection: only firms with certain (often unobserved) competencies tend to trade in international markets. If the gap in unobserved competencies between trading and non-trading firms is larger than the gap in costs of bank intermediation —since banks face imperfect information—then estimates based solely on trading firms understate the role of these underlying competencies in shaping the use of letters of credit.

The role of firm size illustrates this dynamic particularly well. Among firms that already trade internationally, the difference in trade-related competencies between small and large firms is relatively narrow: in other words, small importers tend to be capable and experienced. If banks imperfectly observe these competencies, they may offer less favorable terms to smaller firms, leading to a cost advantage for larger firms. This translates into a positive association between firm size and the use of letters of credit in regressions limited to trading firms. However, once we control for selection (as in column 3), a different picture emerges. Across the broader population, the

¹¹At the end of 2022, the combined total assets of state-owned commercial banks and joint-stock commercial banks accounted for 84.6 percent of Viet Nam’s banking system’s total assets according to the IMF ([International Monetary Fund, 2024](#))

competency gap between small and large firms is wider, and large firms may be better able to access alternative financing. As a result, we observe a reversal in the coefficient: within the full firm population, larger firms may rely less on letters of credit than smaller ones; an insight that is particularly relevant for first-time trading firms or potential future traders.

This pattern holds for other firm characteristics as well. The competency gaps between older, more productive, foreign-owned, or private firms and their counterparts tend to narrow when we look only at firms that engage in international trade. These unobserved competencies—key to successfully navigating foreign markets—can reduce the need for letters of credit. After correcting for selection in column 3, the positive coefficients on age and revenue per worker diminish in size and significance. Meanwhile, the negative coefficient on foreign ownership nearly doubles in magnitude. As expected, private firms are also less likely than state-owned firms to use letters of credit.

Results are notably different in column 4, which controls for all time-invariant firm-level variation through firm fixed effects. In this specification, which only exploits variation within firms over time, only labor productivity, prior import experience, and transaction characteristics remain highly significant.

Letters of credit for exports. Next, consider results on the use of letters of credit for exports shown in Table 4.

Exporters are more likely to use letters of credit if they are older, domestically owned, and more productive. Quantitatively, foreign ownership status has the largest effect, reducing the probability of letter of credit use by over 2 to almost 5 percent, followed by labor productivity. For exports, controlling for selection by adding the Inverse Mills Ratio shows similar effects on the use of letters of credit to imports: domestically owned firms and smaller firms are more likely to rely on letter of credit when benchmarked against the entire firm population rather than only exporting firms. Among trading firms, while prior export experience is not statistically significant, larger values of

Table 4: Linear Probability Results: Exporters 2016-2022

	(1)	(2)	(3)	(4)
Age	0.177*** (0.039)	0.177*** (0.039)	0.181*** (0.059)	
Foreign ownership	-2.719*** (0.337)	-2.784*** (0.338)	-4.760*** (1.102)	-0.706 (0.642)
State ownership	-0.469 (1.455)	-0.590 (1.447)	-0.955 (1.382)	4.842 (4.304)
Employment (log)	0.292** (0.114)	0.093 (0.115)	-0.654* (0.355)	0.337*** (0.127)
Revenue per worker	1.280*** (0.122)	1.088*** (0.121)	0.691*** (0.225)	0.351*** (0.092)
Exported for 1+ years	-0.034 (0.267)	-0.383 (0.270)	-0.652* (0.375)	-0.083 (0.189)
Number of country-products exported	0.002 (0.006)	0.003 (0.006)	0.004 (0.012)	-0.001* (0.001)
Value of transactions (log Exports)		0.570*** (0.041)	0.556*** (0.050)	0.430*** (0.025)
Number of transaction months (exports)		0.154*** (0.032)	0.175*** (0.029)	0.286*** (0.019)
Inverse Mills Ratio			-3.781** (1.568)	
Product-Partner-Year FE	Yes	Yes	Yes	Yes
Sector-Year FE	Yes	Yes	Yes	Yes
Province-Year FE	Yes	Yes	Yes	Yes
Firm FE	No	No	No	Yes
R ²	0.30	0.31	0.31	0.51
Observations	1,045,895	1,045,895	925,465	1,039,413

Source: Matched Customs and Industrial Census (VES) data.

Note: The sample retained for these statistics includes transactions of firms with at least 5 employees, aggregated at the firm-product-partner-year level. The dependent variable on letter-of-credit use is binary, rescaled to 100 for positive use. Errors in all specifications are clustered at the firm level. Specification (4) errors are bootstrapped following the introduction of Inverse Mills Ratio as a regressor. Standard errors in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels respectively.

exports and a higher frequency of transactions throughout the year are associated with greater use of letters of credit. Contrary to importers, state owned exporters or firms with a wider-ranging export portfolio do not differ systematically in their use of letters of credit from other firms.

To summarize, firm characteristics affect letter of credit use more strongly for imports than for exports, which is consistent with our variance decomposition exercises that also suggest a larger role of firm heterogeneity for imports than for exports. Importers, after all, typically bear the cost of issuing letters of credit, while exporters the cost of confirmation in case of high transaction risk.

4.3.1 Which firm characteristics make the greatest difference?

Standardizing all non-binary variables to have zero mean and unit standard deviation allows us to compare the impact of relative differences along the distribution of firm characteristics on the probability of using a letter of credit (see Table 5, where columns (1) and (2) of the main specification are reproduced with standardized variables).

With the exception of age, variation in firm characteristics generally has a greater impact on the probability of using a letter of credit for imports than for exports. Among importers, ownership characteristics stand out with foreign ownership decreasing the likelihood of using a letter of credit by about 7 percentage points, while state ownership increases it by 6 percentage points. Prior import experience has a smaller effect, associated with a 1–2 percentage point reduction in letter of credit usage. This effect is more pronounced than the effect of a one standard deviation increase in firm age. A one standard deviation increase in firm size and productivity increases the probability of using a letter of credit between 1.6–2.2 percentage points, whereas either the value or the number of transactions per partner-product pair has a smaller effect of less than one percentage point.

Consistent with the main specification results in Table 4, firm age, foreign ownership, and revenue per worker are the most relevant factors for letter of credit use in exports. Specifically, foreign ownership is associated with a 2.7 percentage point reduction, a one standard deviation increase in age and productivity with an about 1.5 percentage point increase, and transaction-level variables with an impact of around half a percentage point.

Table 5: Standardized Linear Probability Results: 2016-2022

	Imports		Exports	
	(1)	(2)	(3)	(4)
Age (std)	0.561*** (0.215)	0.565*** (0.215)	1.629*** (0.378)	1.633*** (0.381)
Foreign ownership	-6.981*** (0.258)	-6.676*** (0.263)	-2.725*** (0.336)	-2.790*** (0.338)
State ownership	6.092*** (1.940)	6.134*** (1.950)	-0.458 (1.459)	-0.575 (1.452)
Employment (log) (std)	2.251*** (0.154)	1.839*** (0.151)	0.505** (0.247)	0.115 (0.251)
Revenue per worker (std)	1.960*** (0.097)	1.626*** (0.094)	1.650*** (0.180)	1.391*** (0.179)
Imported or Exported for 1+ years	-1.730*** (0.186)	-1.488*** (0.183)	-0.091 (0.284)	-0.442 (0.287)
Number of country-products (Import or Export) (std)	-1.848*** (0.174)	-1.311*** (0.167)	0.581 (1.502)	0.992 (1.509)
Value of transactions (log Imports or Exports) (std)		2.641*** (0.065)		1.890*** (0.111)
Number of Import or Export transactions (std)		-0.834*** (0.082)		0.529*** (0.101)
Product-Partner-Year FE	Yes	Yes	Yes	Yes
Sector-Year FE	Yes	Yes	Yes	Yes
Province-Year FE	Yes	Yes	Yes	Yes
R ²	0.26	0.26	0.30	0.31
Observations	3,320,731	3,320,731	1,045,895	1,045,895

Source: Matched Customs and Industrial Census (VES) data.

Note: The sample retained for these statistics includes transactions of firms with at least 5 employees, aggregated at the firm-product-partner-year level. The dependent variable for letter-of-credit use is binary, rescaled to 100 for positive use. Errors in all specifications are clustered at the firm level. Standard errors in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels respectively.

4.4 Drivers of Heterogeneous Effects across Firms

This section extends the empirical analysis of heterogeneous firms across different origins or destinations countries, testing directly the second prediction in proposition 2. We then provide additional evidence on firm characteristics associated with cost advantages, testing a foundational hypothesis

motivating our earlier analysis.

4.4.1 Firm characteristics and country-level risk.

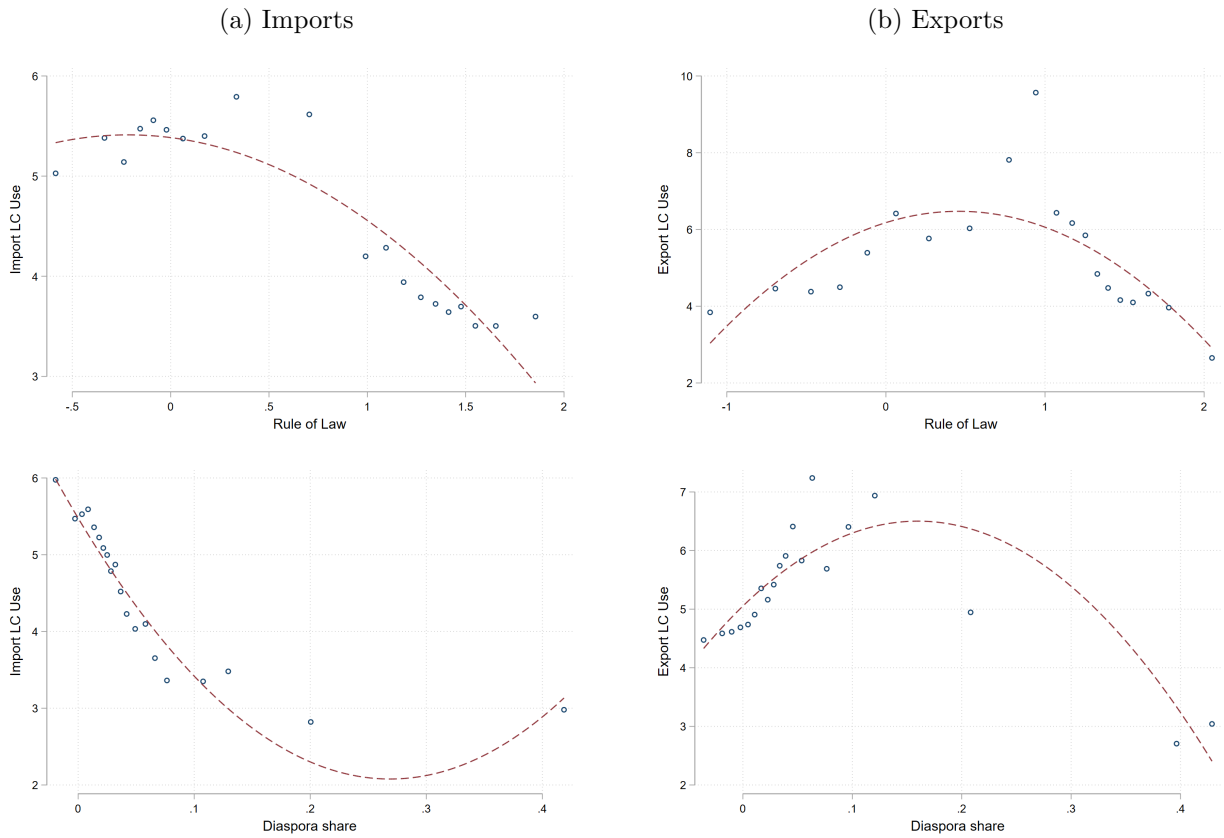
The previous literature has emphasized the central role of country-level characteristics, and in particular contract enforcement for payment choice in international trade (e.g., [Schmidt-Eisenlohr, 2013](#); [Antras and Foley, 2015](#)). Enforcement directly affects the probability that the importer pays and the exporter delivers the goods and also affects the cost of a letter of credit ([Niepmann and Schmidt-Eisenlohr, 2017a](#)), with costs rising with the risk of the destination country. Descriptive statistics in our sample confirm that letters of credit are used less to trade with markets with high rule of law, in what appears to be an inverse U-shaped relationship (see Figure 4 top panels). As contract enforcement worsens, there is higher demand for letters of credit, but not when rule of law collapses.

Firm heterogeneity may interact more broadly with country risk and specifically the availability of market information about a partner country. When more information is available about a foreign market, this improves the assessment of the reliability of potential partners, lowers risk, and reduces transaction and information costs for all firms independent of their screening ability. Firms hence need to rely less on letters of credit in their transactions. However, these gains are unlikely to be the same across the distribution of firms with different characteristics. Proposition 2 formalizes the mechanism behind the hypothesis: better partner-country features attenuate the effect of cost and screening advantages enjoyed by firms of different characteristics. In both cases, improvements in partner-country information availability or institutional quality compress heterogeneity in letter of credit use across firms.

On the one hand, firms that are better at screening should reduce their reliance on letters of credit by *less* than firms that are worse at screening, when market information or institutional quality improves. The reason is that firms with weaker screening abilities are expected to depend

more heavily on letters of credit precisely in destinations where information is scarce or where their contract is not easily enforced. On the other hand, firms with cheaper access to bank intermediation should reduce their use of letters of credit by *more* than firms without cost advantages. This could be the case because these firms' cost advantage and access to letters of credit is expected to be particularly strong in the riskiest and lowest information countries.

Figure 4: Use of letters of credit by partner country's characteristics



Source: Matched Customs and Industrial Census (VES) data. Rule of Law Index comes from the World Bank Global Governance Indicators. Diaspora data comes from UN Statistics Division (UNSD) Demographic Statistics Database. *Note:* Bin scatters control for HS4 product and year fixed effects. Rule of Law estimates are standardized globally at the year level, with zero corresponding to the global average and 1 to standard deviations from that average. Diaspora shares are calculated at the year level based on a 2020 cross-section share of Vietnamese migrant stock in partner country relative to the total stock.

To test these hypotheses, we interact firm characteristics with two country characteristics: the

rule of law index and the Vietnamese diaspora share. The rule of law index has been used extensively to capture the strength of contract enforcement institutions in previous studies, including to explain cross-country trade finance flows (Niepmann and Schmidt-Eisenlohr, 2017b). However, one caveat is that Vietnamese traders may have closer relationships and hence better information in markets with lower institutional quality. To address this caveat, we use diaspora share as a proxy for the ease of obtaining market information about that partner country. The measure of migrant stock has been used extensively in the trade literature to identify countries with lower information frictions fostering trade (Gould, 1994; Head and Ries, 1998; Felbermayr and Toubal, 2012). Similarly to rule of law, the descriptive statistics show that letters of credit are used less when trading with markets that have high diaspora shares (see the bottom panels of Figure 4). As market information becomes scarcer, demand for letters of credit rises. For exports, we observe a similar inverse-U shaped pattern: letter-of-credit use declines again for destinations at the lower tail of the diaspora-share distribution.

We test for these relationships formally with the following difference-in-differences specification

$$LC_{ipjt} = \beta_0 + \beta_1 \mathbf{X}_{it} + \beta_2 (\mathbf{X}_{it} \times D_j) + \beta_3 \mathbf{Z}_{ipjt} + \beta_4 (\mathbf{Z}_{ipjt} \times D_j) + \gamma_{pjt} + \phi_{rt} + \lambda_{st} + \varepsilon_{ipjt} \quad (9)$$

where LC_{ipjt} is the probability of firm i using a letter of credit for trade of product p with country j at year t ; $\mathbf{X}_{it}$ is a vector of firm i characteristics at year t including state or foreign ownership, age, employment, productivity, and prior trade experience; $\mathbf{Z}_{ipjt}$ is a vector of the firm's trade activity such as dollar volume and number of customs transactions for product p with country j ; D_j is a measure of partner country's rule of law or Vietnamese diaspora share present in that country; γ_{pjt} , ϕ_{rt} , and λ_{st} are product-partner-year fixed effects, province-year fixed effects, and sector-year fixed effects, respectively.

Table 6 confirms the predictions in proposition 2 that stronger contract enforcement and more

Table 6: Linear Probability interactions with rule of law index and diaspora shares: 2016-2022

	Rule of Law			Diaspora		
	Imports		Exports	Imports		Exports
	Main	Interaction	Main	Interaction	Main	Interaction
Age (std)	0.820*** (0.292)	-0.389*** (0.134)	1.788*** (0.321)	-0.194** (0.0866)	0.673*** (0.257)	-1.714** (0.721)
Foreign ownership	-7.249*** (0.322)	0.837*** (0.181)	-2.633*** (0.355)	-0.209 (0.214)	-7.193*** (0.302)	8.176*** (0.954)
State ownership	7.455*** (2.396)	-1.822* (1.082)	0.883 (1.635)	-1.574 (0.987)	6.538*** (2.241)	-0.624 (5.199)
Employment (log) (std)	2.318*** (0.199)	-0.689*** (0.125)	-0.282 (0.433)	0.383 (0.247)	2.112*** (0.177)	-4.036*** (0.653)
Productivity (std)	1.770*** (0.110)	-0.247*** (0.0675)	1.239*** (0.241)	0.153 (0.111)	1.780*** (0.105)	-2.213*** (0.399)
Trade experience (1+ yrs)	-1.726*** (0.222)	0.228 (0.149)	-0.328 (0.281)	-0.129 (0.196)	-1.724*** (0.215)	2.847*** (1.068)
Number of country-products traded (std)	-1.609*** (0.197)	0.436*** (0.0876)	2.734 (2.901)	-1.473 (1.205)	-1.487*** (0.183)	1.989*** (0.447)
Trade value (std)	2.812*** (0.0856)	-0.280*** (0.0559)	2.310*** (0.161)	-0.419*** (0.154)	2.867*** (0.0786)	-2.894*** (0.336)
Transaction months (std)	-0.700*** (0.113)	-0.195** (0.0760)	0.379*** (0.106)	0.161** (0.0788)	-0.842*** (0.106)	-0.264 (0.458)
Constant	8.161*** (0.240)		8.071*** (0.807)		8.014*** (0.245)	7.881*** (0.597)
Product-year FE	Yes	Yes	Yes	Yes	Yes	Yes
Sector-year FE	Yes	Yes	Yes	Yes	Yes	Yes
Province-year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	3,312,437		1,044,688		3,074,484	857,826
R ²	0.264		0.307		0.239	0.299

Source: Matched Customs and Industrial Census (VES) data. UN Statistics Division (UNSD) Demographic Statistics Database.

Note: The sample retained for these statistics includes transactions of firms with at least 5 employees, aggregated at the firm-product-partner-year level. Rule of Law estimates are standardized globally at the year level, with zero corresponding to the global average and 1 to standard deviations from that average. Diaspora shares are calculated at the year level based on a 2020 cross-section share of Vietnamese migrant stock in partner country relative to the total stock. The dependent variable for letter-of-credit use is binary, rescaled to 100 for positive use. Standard errors in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels respectively.

information availability in the partner country should attenuate the effects of firm characteristics. That is, as predicted by the theory, the interaction terms for the different firm characteristics have the opposite signs from their baseline effects.

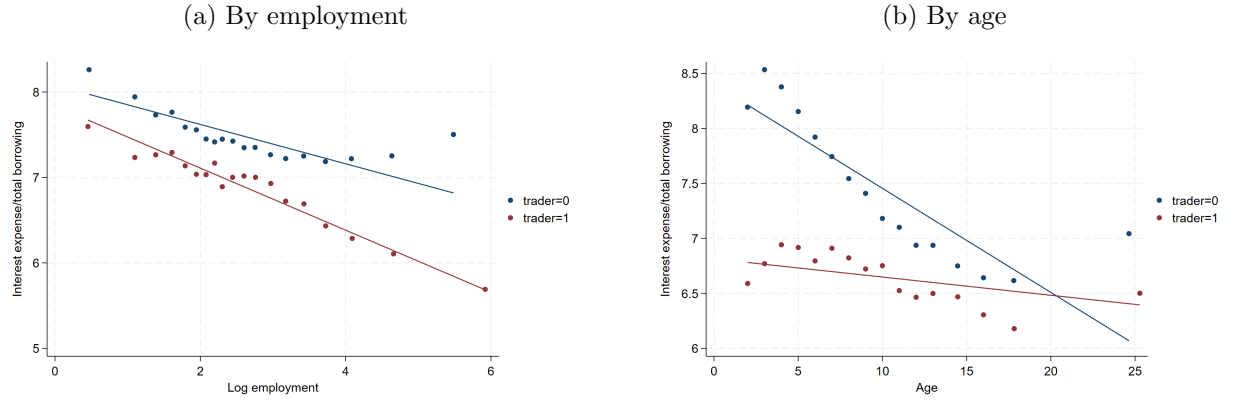
Among importers, characteristics where the cost channel dominates (age, firm size, and productivity) have positive main effects and negative interactions with rule of law or diaspora share. In other words, their cost advantage in accessing letters of credit shrinks when partner country information availability increases. Characteristics that primarily improve screening (foreign ownership and import experience) have negative main effects and positive interactions. That is, their screening advantage matters less when trading partners' institution quality or information improves. In every case where both the main effect and interaction are statistically significant, the signs are opposite, as predicted. Results for exporters follow the same pattern for foreign ownership (screening) and age and productivity (cost), though fewer variables are significant. This is consistent with our earlier finding that firm heterogeneity matters more for letter of credit for imports than exports. Overall, letter of credit use becomes flatter with respect to firm characteristics as contract enforcement or information improves, consistent with a general reduction in dependence on bank intermediation.

4.4.2 Cost of bank intermediation.

A key driver of letter of credit use is the fees charged by banks. As discussed above, these fees can vary with destination country risk ([Niepmann and Schmidt-Eisenlohr, 2017a](#)). However, they can also depend on firm-specific characteristics that affect firms' risk profiles. While we do not observe letter of credit fees directly, we can examine how the general cost of banking intermediation varies with firm characteristics. It seems plausible that results on differential pricing across firm characteristics would extend to the pricing of trade finance.

More specifically, we use VES data to construct an effective interest rate proxy for firms that

Figure 5: Interest rate by trading status across firm sizes and age



Source: Customs and VES data.

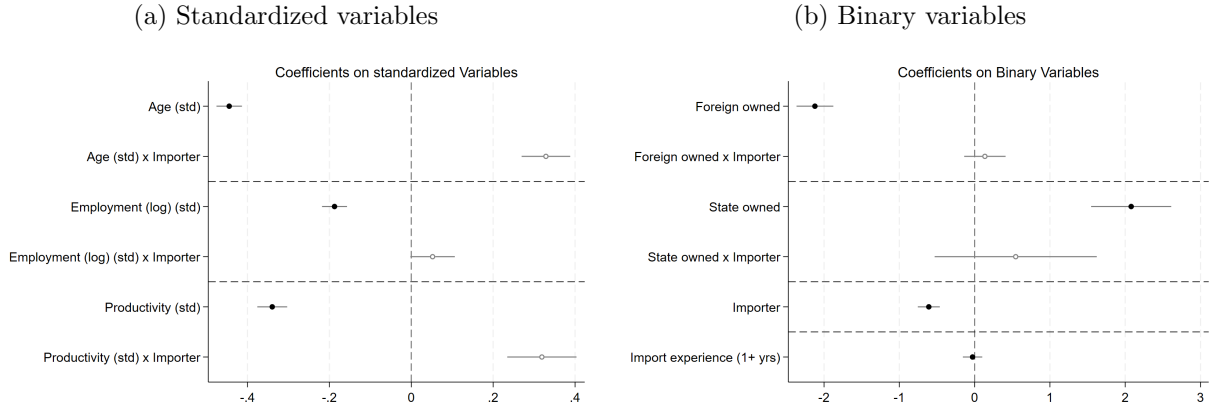
Note: Interest rate is calculated as the ratio of interest expense over total long-term and short-term debt, averaged between the beginning and end of the year. The sample only includes firms with at least 5 employees.

reported borrowing figures as well as interest expenses on their balance sheet. The effective interest rate proxy is calculated as the ratio of interest expense over total long-term and short-term debt, averaged between the beginning and end of the year. Figure 5 shows the average estimated interest rate along the distribution of firm age and employment. Overall, it appears that lenders consider trading firms lower risk: at all ages and sizes, trading firms reported effectively lower interest rate than non-trading firms. Smaller and younger firms appear to be considered riskier: for both traders and non-traders, effective interest rate decreases with firm size and age. However, the relationship between age and interest charges is flatter for trading firms.

Trade status not only shifts the cost curves but may also alter their slope, indicating that the gaps in effective interest rates across the firm size or age distribution can differ between traders and non-traders. Regressing average interest rates on firm characteristics and their interactions with trading status provides additional insights. Our analysis focuses on importers, as the cost of letters of credit is borne primarily by the importer; while processing and negotiating costs for exporters are negligible in comparison.

Figure 6 illustrates how interest rates vary with firm characteristics in a specification that

Figure 6: Importer characteristics and interest rate (2016-2022)



Source: Customs and VES data.

Note: Coefficient estimates from regressing interest rate on firm characteristics, sector-year and province-year fixed effects. Errors are clustered at the firm level. Interest rate is calculated as the ratio of interest expense over total long-term and short-term debt, averaged between the beginning and end of the year. The sample only includes firms with at least 5 employees over the entire census.

includes all characteristics jointly. Controlling for sector-year and province-year fixed effects, we find that older, larger, foreign-owned, more productive firms, as well as importers, enjoy significantly lower effective interest charges on average across the full firm population.

Interactions between firm characteristics and import status reveal that the advantages associated with age and productivity are less pronounced among trading firms compared to non-trading firms, as also shown in Figure 5. For foreign-owned traders and larger trading firms, the interaction effects are not statistically significant, suggesting a level shift in the cost curve associated with imports rather than a change in its slope. Put differently, the relative cost differential in bank intermediation between large and small firms remains roughly the same for trading firms. This supports the hypothesis that banks incorporate competency differences between firms imperfectly, if at all, into the effective interest they charge (see discussion on selection in Section 4.3). While import status is associated with somewhat lower charges, import experience is not statistically significant, reinforcing the same conclusion. State ownership, on the other hand, is positively correlated with effective interest rates regardless of trading status, possibly reflecting particularities

in the relationship between banks and state-owned firms in the Vietnamese context (see earlier discussion of Main Results).

Limited evidence from a 2023 IFC survey of trading firms in Viet Nam, which also reported prices of letters of credit and were matched through their tax identification numbers, suggests a positive correlation between the estimated effective interest rate and the reported letter of credit cost (as a share of the transaction value).

This evidence is consistent with the model where higher letter of credit use for older and larger firms is driven in part by their lower cost of obtaining letters of credit. For foreign owned firms and experienced importers, despite lower cost of financial intermediation, the relative gain from letter of credit use is still small to translate into a higher probability of use.

5 Conclusion

Letters of credit are an important tool to facilitate trade, especially for developing and emerging economies. Despite the large interest in this instrument among policy makers, little is known about the firm-level determinants of letter of credit use. This study fills this gap by providing first estimates of these relationships.

There are three main takeaways for policymakers from this study. First, we show that firm-level heterogeneity is not only relevant but accounts for a much larger share of the variation in letter of credit use than country-product factors. Development initiatives aimed at expanding the availability of letters of credit to a wider range of markets and banks need to be complemented by even greater efforts to improve access for firms that are systematically in greater need—such as those lagging behind in terms of relevant competencies or facing cost disadvantages by existing banks. Second, while national development policies often focus on exporters and multinational firms, our evidence highlights the disproportionate gaps in letters of credit use among domestic importing firms, whose access to such instruments can support more broadly the internationalization

of developing economies. Third, our results suggest that younger, smaller, and first-time traders stand to benefit from interventions to improve access to letters of credit. Addressing this need requires not only working with the financial sector to better assess and manage firms' risks, but also expanding advisory services to help firms navigate the complexities of foreign markets. Letters of credit provide needed security; yet, to some extent and in some markets, may substitute for capabilities that firms can develop internally and, moreover, share across their networks.

Although Viet Nam provides a quite diverse and international market to test these hypotheses, future work using more detailed datasets across countries and in contexts of market fragmentation or policy shocks would further enrich the literature. Better understanding banks' heterogeneous attitudes towards risk in cross-border transactions, and pricing of trade finance instruments like letters of credit, would support the effectiveness of development interventions in this area.

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A Theory Appendix

This appendix provides proofs for the two propositions stated in the main text.

A.1 Proof of Proposition 1 (Letter of Credit Determinants)

Proof. Recall from equation (1) that the expected net gain from using a letter of credit relative to the best alternative is

$$E[\Delta\Pi] = \frac{R - (1 + r^I)m}{(1 + r^E)(1 + f(1 + r^I))} - \max \left\{ \frac{\tilde{\eta}^I R}{1 + r^E}, \frac{\tilde{\eta}^E R}{1 + r^I} \right\}.$$

Throughout, we maintain the feasibility condition $R > (1 + r^I)m$, which is necessary for the letter of credit payment $P^{LC} = \frac{R - (1 + r^I)m}{1 + f(1 + r^I)}$ to be positive.

Part (i): Monitoring cost m and variable cost f .

The alternative profit $E[\Pi^{Alt}]$ is independent of m and f . Hence:

$$\frac{\partial E[\Delta\Pi]}{\partial m} = \frac{-(1 + r^I)}{(1 + r^E)(1 + f(1 + r^I))} < 0.$$

For the variable cost:

$$\frac{\partial E[\Delta\Pi]}{\partial f} = \frac{-(1 + r^I)[R - (1 + r^I)m]}{(1 + r^E)[1 + f(1 + r^I)]^2} < 0,$$

where the inequality follows from $R > (1 + r^I)m$.

Part (ii): Probability of direct payment or delivery ($\tilde{\eta}^I$ or $\tilde{\eta}^E$).

The letter of credit profit $E[\Pi^{LC}]$ does not depend on $\tilde{\eta}^I$ or $\tilde{\eta}^E$, since the bank resolves all

commitment problems. In contrast, $E[\Pi^{Alt}]$ is weakly increasing in both $\tilde{\eta}^I$ and $\tilde{\eta}^E$. Specifically:

$$\frac{\partial E[\Pi^{Alt}]}{\partial \tilde{\eta}^I} = \begin{cases} \frac{R}{1+r^E} > 0 & \text{if trade credit is the best alternative} \\ 0 & \text{if cash in advance is the best alternative,} \end{cases}$$

and

$$\frac{\partial E[\Pi^{Alt}]}{\partial \tilde{\eta}^E} = \begin{cases} 0 & \text{if trade credit is the best alternative} \\ \frac{R}{1+r^I} > 0 & \text{if cash in advance is the best alternative.} \end{cases}$$

In both cases:

$$\frac{\partial E[\Delta\Pi]}{\partial \tilde{\eta}^j} = -\frac{\partial E[\Pi^{Alt}]}{\partial \tilde{\eta}^j} \leq 0,$$

with strict inequality when the binding alternative involves $\tilde{\eta}^j$. Higher reliability of the counterparty makes the non-intermediated alternative more profitable, reducing the relative attractiveness of the letter of credit.

Part (iii): Revenues R .

To match the statement in the proposition, consider the expected net gain *per unit of transaction value*. Since $R > 0$, we can write

$$\frac{E[\Delta\Pi]}{R} = \frac{1}{(1+r^E)(1+f^{LC}(1+r^I))} - \max\left\{\frac{\tilde{\eta}^I}{1+r^E}, \frac{\tilde{\eta}^E}{1+r^I}\right\} - \frac{(1+r^I)m}{(1+r^E)(1+f^{LC}(1+r^I))} \cdot \frac{1}{R}.$$

The first two terms are independent of R . Hence,

$$\frac{\partial}{\partial R} \left(\frac{E[\Delta\Pi]}{R} \right) = \frac{(1+r^I)m}{(1+r^E)(1+f^{LC}(1+r^I))} \cdot \frac{1}{R^2} > 0.$$

Therefore, the expected net gain from using a letter of credit, measured per unit of transaction value, increases with revenues R . Intuitively, a larger transaction allows the fixed monitoring cost

m to be spread over a greater value, making the letter of credit relatively more attractive. $\square$

A.2 Proof of Proposition 2 (Firm Characteristics and Letter of Credit Use)

Proof. Assume the monitoring cost is given by

$$m(\theta^E, \theta^I, \phi^E, \phi^I) = \bar{m}[(1 - \theta^E)(1 - \phi^I) + (1 - \theta^I)(1 - \phi^E)],$$

that $\eta^j = \theta^i$ so that $\tilde{\eta}^j = \theta^i + (1 - \theta^i)\phi^j$ for $i, j \in \{I, E\}$ with $i \neq j$, and that $f = 0$. The expected net gain is

$$E[\Delta\Pi] = \frac{R - (1 + r^I)m}{1 + r^E} - \max \left\{ \frac{\tilde{\eta}^I R}{1 + r^E}, \frac{\tilde{\eta}^E R}{1 + r^I} \right\}. \quad (10)$$

We prove the result for exporter firm quality θ^E ; the case for importer firm quality θ^I follows by a symmetric argument (detailed at the end of this proof).

Key partial derivatives. By the cross-side structure of the monitoring cost:

$$\frac{\partial m}{\partial \theta^E} = -\bar{m}(1 - \phi^I), \quad (11)$$

$$\frac{\partial \tilde{\eta}^I}{\partial \theta^E} = 1 - \phi^I, \quad (12)$$

$$\frac{\partial \tilde{\eta}^E}{\partial \theta^E} = 0. \quad (13)$$

Equation (11) holds because only the first term of m depends on θ^E . Equation (12) follows from $\tilde{\eta}^I = \theta^E + (1 - \theta^E)\phi^I$. Equation (13) holds because $\tilde{\eta}^E = \theta^I + (1 - \theta^I)\phi^E$ does not involve θ^E .

Part (i): Main effect of θ^E .

Case 1: Trade credit is the best alternative, i.e., $\frac{\tilde{\eta}^I R}{1 + r^E} \geq \frac{\tilde{\eta}^E R}{1 + r^I}$. Then:

$$E[\Delta\Pi] = \frac{R - (1 + r^I)m - \tilde{\eta}^I R}{1 + r^E}.$$

Using (11) and (12):

$$\begin{aligned}\frac{\partial E[\Delta\Pi]}{\partial\theta^E} &= \frac{1}{1+r^E} [(1+r^I)\bar{m}(1-\phi^I) - (1-\phi^I)R] \\ &= \frac{(1-\phi^I)}{1+r^E} [(1+r^I)\bar{m} - R].\end{aligned}\tag{14}$$

The first term in the bracket, $(1+r^I)\bar{m}$, is the *cost channel*: higher θ^E reduces monitoring costs, making the letter of credit more attractive. The second term, R , is the *screening channel*: higher θ^E raises $\tilde{\eta}^I$, making trade credit safer and thus reducing the benefit of the letter of credit. The sign depends on which channel dominates.

Case 2: Cash in advance is the best alternative, i.e., $\frac{\tilde{\eta}^E R}{1+r^I} > \frac{\tilde{\eta}^I R}{1+r^E}$. Then:

$$E[\Delta\Pi] = \frac{R - (1+r^I)m}{1+r^E} - \frac{\tilde{\eta}^E R}{1+r^I}.$$

Since $\tilde{\eta}^E$ does not depend on θ^E (equation (13)), only the cost channel operates:

$$\frac{\partial E[\Delta\Pi]}{\partial\theta^E} = \frac{(1+r^I)\bar{m}(1-\phi^I)}{1+r^E} > 0.\tag{15}$$

The exporter's screening ability is irrelevant for the riskiness of cash in advance (where the importer pays first and the delivery risk depends on the *exporter's* reliability $\tilde{\eta}^E$, driven by θ^I), so only the cost reduction matters.

Summary. If the cost channel dominates—i.e., the reduction in bank monitoring costs exceeds the improvement in screening—then $\partial E[\Delta\Pi]/\partial\theta^E > 0$ (Cost Dominance). If the screening channel dominates, then $\partial E[\Delta\Pi]/\partial\theta^E < 0$ (Screening Dominance). Under cash in advance, only the cost channel is active, so the effect is unambiguously positive.

Part (ii): Cross-derivative with partner-country institutions.

In both cases, the main effect of θ^E is proportional to $(1-\phi^I)$. Taking the cross-derivative with

respect to ϕ^I :

Case 1 (Trade credit): From (14),

$$\frac{\partial^2 E[\Delta\Pi]}{\partial\theta^E \partial\phi^I} = \frac{R - (1 + r^I)\bar{m}}{1 + r^E}.$$

If cost dominance holds $[(1 + r^I)\bar{m} > R]$, so main effect is positive], then this cross-derivative is negative. If screening dominance holds $[(1 + r^I)\bar{m} < R]$, so main effect is negative], then this cross-derivative is positive. In both cases, **the cross-derivative has the opposite sign of the main effect**.

Case 2 (Cash in advance): From (15),

$$\frac{\partial^2 E[\Delta\Pi]}{\partial\theta^E \partial\phi^I} = -\frac{(1 + r^I)\bar{m}}{1 + r^E} < 0.$$

The main effect (15) is positive, and the cross-derivative is negative. Again, **opposite signs**.

Part (iii): Convergence as $\phi^I \rightarrow 1$.

In both cases, the main effect is proportional to $(1 - \phi^I)$. As $\phi^I \rightarrow 1$:

$$\frac{\partial E[\Delta\Pi]}{\partial\theta^E} \rightarrow 0.$$

The intuition has two parts. First, when partner-country institutions are perfect ($\phi^I = 1$), the alternative payment term (trade credit) is risk-free regardless of the exporter's screening ability ($\tilde{\eta}^I = \theta^E + (1 - \theta^E) \cdot 1 = 1$), so there is no screening advantage to compress. Second, monitoring costs on the exporter side vanish (m includes the factor $(1 - \theta^E)(1 - \phi^I) = 0$), so there is no cost advantage to exploit. Both channels shut down, and firm heterogeneity in θ^E ceases to affect letter of credit use.

Symmetric argument for importer firm quality θ^I .

The relevant partial derivatives are:

$$\begin{aligned}\frac{\partial m}{\partial \theta^I} &= -\bar{m}(1 - \phi^E), \\ \frac{\partial \tilde{\eta}^E}{\partial \theta^I} &= 1 - \phi^E, \\ \frac{\partial \tilde{\eta}^I}{\partial \theta^I} &= 0.\end{aligned}$$

Now the roles reverse across regimes.

Case 1 (Trade credit): Since $\tilde{\eta}^I$ does not depend on θ^I , only the cost channel operates:

$$\frac{\partial E[\Delta\Pi]}{\partial \theta^I} = \frac{(1 + r^I)\bar{m}(1 - \phi^E)}{1 + r^E} > 0.$$

Case 2 (Cash in advance): Both channels are active:

$$\frac{\partial E[\Delta\Pi]}{\partial \theta^I} = \frac{(1 + r^I)\bar{m}(1 - \phi^E)}{1 + r^E} - \frac{(1 - \phi^E)R}{1 + r^I} = (1 - \phi^E) \left[\frac{(1 + r^I)\bar{m}}{1 + r^E} - \frac{R}{1 + r^I} \right].$$

The sign is ambiguous, depending on whether cost or screening dominates.

In both cases, the main effect is proportional to $(1 - \phi^E)$. The cross-derivative with respect to ϕ^E has the opposite sign of the main effect:

$$\begin{aligned}\frac{\partial^2 E[\Delta\Pi]}{\partial \theta^I \partial \phi^E} &= -\frac{(1 + r^I)\bar{m}}{1 + r^E} < 0 \quad (\text{Case 1, main effect positive}), \\ \frac{\partial^2 E[\Delta\Pi]}{\partial \theta^I \partial \phi^E} &= -\left[\frac{(1 + r^I)\bar{m}}{1 + r^E} - \frac{R}{1 + r^I} \right] \quad (\text{Case 2, opposite sign of main effect}).\end{aligned}$$

As $\phi^E \rightarrow 1$, the factor $(1 - \phi^E) \rightarrow 0$, so $\partial E[\Delta\Pi]/\partial \theta^I \rightarrow 0$, completing the proof. $\square$

B Appendix: Robustness Tests

Share of trade covered by letters of credit as dependent variable. In our baseline results, we create a dummy variable that equals one if a firm uses any letters of credit for a product-market combination in a given year. Alternatively, we can calculate the share of transactions in a month that rely on letters of credit for a firm-product-market and use that as our dependent variable. The share may be a more appropriate dependent variable if firms vary their use of letters of credit at the product-partner-year level, eventually covering only part of transactions at that level.

Table 7: Robustness check: using share of transactions covered by LC

	Imports (1)	Exports (2)
Firm age	0.055** (0.022)	0.175*** (0.039)
Foreign ownership	-5.875*** (0.231)	-1.850*** (0.264)
State ownership	5.768*** (1.865)	-0.032 (1.283)
Employment (log)	0.831*** (0.071)	-0.084 (0.100)
Revenue per worker	1.066*** (0.064)	0.679*** (0.105)
Imported or Exported for 1+ years	-1.338*** (0.163)	-0.430* (0.246)
Number of country-products imported	-0.006*** (0.001)	0.005 (0.006)
Value of transactions (log Imports or exports)	0.821*** (0.020)	0.461*** (0.038)
Number of transaction months (imports or exports)	-0.408*** (0.023)	-0.190*** (0.028)
Constant	-6.392*** (0.488)	-3.385*** (0.820)
Product-partner-year FE	Yes	Yes
Sector-year FE	Yes	Yes
Province-year FE	Yes	Yes
R ²	0.27	0.31
Observations	3,320,731	1,045,895

Note: Matched Customs and Industrial Census (VES) data. The sample includes firm-product-partner-year level observations for firms with at least 5 employees. Dependent variable is the share of transaction value using letters of credit (LC). Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Results are reported in table 7. Coefficient estimates for firm age, foreign ownership, productivity, and employment are similar, likely because most firms in the data report no more than a single transaction per product-partner-year. However, we do see that the number of transactions has a negative effect on the share of letters of credit used.